The Information Complexity of Nash Equilibrium

N. Minimax

Question

Let $\mathcal{M}$ be the class of matrices in $[-1,1]^{3\times 3}$ having a unique, completely mixed equilibrium. Querying entry (i,j) produces an independent $N(A_{ij},1)$ observation. Strategies belong to the probability simplex Δ_3. For $\varepsilon>0$ and $0<\delta<1$, write

$$G_A(x,y)=\max_i (Ay)_i-\min_j (x^\top A)_j, \qquad M_{\varepsilon,\delta}(A)=\inf_{\mathcal{A}} \mathbb{E}_A \tau_{\mathcal{A}},$$

where the infimum ranges over sequential procedures satisfying

$$\Pr_B\big(G_B(\hat{x},\hat{y})\le\varepsilon\big)\ge 1-\delta \quad \text{for every } B\in\mathcal{M}.$$

For a fixed $A\in\mathcal{M}\cap(-1,1)^{3\times 3}$, what determines the leading observation complexity as confidence and accuracy increase?

Result

Let (p,q) be the equilibrium of A. Let Δ_9 denote sampling allocations over the nine entries, let Δ_9° denote its strictly positive elements, and let

$$\Pi(p,q)=\{\pi\ge 0:\pi\mathbf{1}=p,\ \pi^\top\mathbf{1}=q\}.$$

For $w\in\Delta_9^\circ$, define

$$c_{kl}^{ij}=\mathbf{1}_{\{k=i\}}q_l-p_k\mathbf{1}_{\{l=j\}}, \qquad \sigma_{ij}(w)=\left(\sum_{kl}\frac{(c_{kl}^{ij})^2}{w_{kl}}\right)^{1/2},$$

and $F_{p,q}(w)=\max_{\pi\in\Pi(p,q)}\sum_{ij}\pi_{ij}\sigma_{ij}(w)$.

Theorem. The following limits exist:

$$\boxed{\lim_{\varepsilon\downarrow 0}\varepsilon^2\lim_{\delta\downarrow 0}\frac{M_{\varepsilon,\delta}(A)}{\log(1/\delta)}=C(p,q):=2\Big[\min_{w\in\Delta_9^\circ}F_{p,q}(w)\Big]^2.} \tag{1}$$

The minimum is attained. Furthermore,

$$0<C(p,q)\le 8, \qquad C(p,q)=8\iff p=q=(1/3,1/3,1/3).$$

Thus the leading complexity depends only on equilibrium probabilities. For

$$A_r=\frac{1}{2}\operatorname{diag}\Big(\frac{r}{1-2r},1,1\Big), \qquad 0<r<1/4,$$

the equilibrium is $p_r=q_r=(1-2r,r,r)$, and $C(p_r,q_r)=\Theta(r^2)$ as $r\downarrow 0$. By comparison, estimating all nine payoffs to maximum-norm error ε, under the same uniform correctness requirement, has coefficient 18 in (1). Payoff estimates with maximum-norm error ε guarantee only a duality gap of 2ε, so a procedure that first estimates the payoffs and then solves the game needs error $\varepsilon/2$ to meet the requirement here; its coefficient is 72.

The limits are ordered as displayed; no uniformity near degeneracy is asserted. All logarithms are natural.

Characterization (2) below applies the infinite-answer framework of [1] to matrix games. Formula (1), the bounds on C, and the example are derived here; we have not established whether they are new.

Proof

Write $z = (x, y)$, $\mathcal{Z}_\varepsilon(A) = \{(x, y) \in \Delta_3 \times \Delta_3 : G_A(x, y) \le \varepsilon\}$, and

$$d_w(A, B) = \frac{1}{2} \sum_{kl} w_{kl}(A_{kl} - B_{kl})^2, \qquad \mathcal{B}_z = \{B \in \mathcal{M} : G_B(z) > \varepsilon\}.$$

Set

$$\Gamma_\varepsilon(A) = \sup_{z \in \mathcal{Z}_\varepsilon(A),\, w \in \Delta_9} \inf_{B \in \mathcal{B}_z} d_w(A, B).$$

For sufficiently small fixed ε, we first establish the Gaussian infinite-answer characterization underlying [1]:

$$\lim_{\delta \downarrow 0} \frac{M_{\varepsilon,\delta}(A)}{\log(1/\delta)} = \Gamma_\varepsilon(A)^{-1}. \tag{2}$$

Its general form is known. We follow the change-of-measure argument of [1] and only sketch its standard steps; the sketch also handles overlapping sets of acceptable answers.

Fix $\alpha > 0$. For each (z, w), choose $B \in \mathcal{B}_z$ with cost below $\Gamma_\varepsilon + \alpha$. Strict invalidity and continuity give a neighborhood on which B remains invalid and its cost is below $\Gamma_\varepsilon + 2\alpha$. Compactness of $\mathcal{Z}_\varepsilon(A) \times \Delta_9$ supplies a finite cover with alternatives $B_1, \dots, B_J$. Partition correct stopped outcomes into events E_j according to their reported answer and realized sampling proportions $N(\tau)/\tau$; assign arbitrary proportions when $\tau = 0$.

Put $s_j = \Pr_A(E_j)$. Since $\Pr_{B_j}(E_j) \le \delta$, conditional Jensen's inequality for the stopped log likelihood L_j gives

$$\mathbb{E}_A[\mathbf{1}_{E_j} L_j] \ge s_j \log(s_j/\delta).$$

Under A,

$$L_j = \tau\, d_{N(\tau)/\tau}(A, B_j) + U_j, \qquad \mathbb{E}_A U_j^2 \le 4\mathbb{E}_A \tau,$$

by Gaussian martingale isometry. Summing and applying Cauchy–Schwarz yields

$$(\Gamma_\varepsilon + 2\alpha)\mathbb{E}_A\tau + 2\sqrt{J\mathbb{E}_A\tau} \ge (1 - \delta)\log(1/\delta) - \log(J + 1). \tag{3}$$

The last term follows by including the discarded incorrect-answer event in the entropy bound. This inequality is uniform over procedures. Letting $\delta \downarrow 0$, then $\alpha \downarrow 0$, proves the lower bound in (2). Procedures with infinite $\mathbb{E}_A\tau$ satisfy that bound trivially.

For achievability, fix z, w with $w > 0$ and $D = \inf_{B \in \mathcal{B}_z} d_w(A, B) > 0$. Positive allocations approximate the supremum: replacing w by $(1 - t)w + t\mathbf{1}/9$ retains at least $1 - t$ of its information. Take $n_{kl} = \lceil n w_{kl} \rceil$, where

$$n = (1 + \rho)\log(1/\delta)/D, \qquad \rho > 0.$$

Using the sample means, define $Q(B) = \frac{1}{2}\sum_{kl} n_{kl}(\bar{A}_{kl} - B_{kl})^2$. For the true matrix B, $Q(B)$ has distribution $\chi^2_9/2$. With

$$L_2 = \log(2/\delta), \qquad b_\delta = L_2 + 3\sqrt{L_2} + 9/2,$$

the Gaussian Chernoff bound gives $\Pr_B(Q(B) > b_\delta) \le \delta/2$. Report z if $\inf_{B \in \mathcal{B}_z} Q(B) > b_\delta$. Otherwise, take $\lceil 8\varepsilon^{-2}\log(36/\delta) \rceil$ fresh observations per entry and return an equilibrium of their sample-mean matrix. A Gaussian union bound makes every payoff error at most $\varepsilon/2$, hence the resulting gap at most ε, except with probability $\delta/2$.

This procedure is uniformly correct. At the target A, the triangle inequality gives

$$\inf_{B \in \mathcal{B}_z} \sqrt{Q(B)} \ge \sqrt{nD} - \sqrt{Q(A)}.$$

Consequently certification succeeds with probability tending to one, and the fallback contributes $o(\log(1/\delta))$ expected observations. Optimizing z, w, then letting $\rho \downarrow 0$, proves (2). These fixed choices are legitimate because the infimum is pointwise; certification ensures correctness throughout $\mathcal{M}$.

We now evaluate its small-accuracy limit. Put $T = \{h : \mathbf{1}^\top h = 0\}$. The maps

$$A : T \longrightarrow p^\perp, \qquad A^\top : T \longrightarrow q^\perp$$

are bijections: a nonzero kernel vector would generate another nearby equilibrium. A neighborhood of A also lies in $\mathcal{M}$. Since A has the unique equilibrium (p, q) and equilibria depend upper hemicontinuously on

payoffs, every game near A has all its optimal strategies near (p, q). Because $p, q > 0$, these strategies are completely mixed, and a game whose optimal strategies are all completely mixed has a unique equilibrium [2].

Define the compact set

$$K = \{(a, b) : p^\top a = q^\top b = 0,\ a_i + b_j \le 1\}.$$

Indeed, the two maxima are nonnegative and sum to at most one; the weighted-zero restrictions then bound every coordinate below. For sufficiently small ε, every point of K corresponds to valid strategies through

$$A(y - q) = \varepsilon a, \qquad -A^\top(x - p) = \varepsilon b, \tag{4}$$

and these are exactly $\mathcal{Z}_\varepsilon(A)$. In particular, $z = (p, q) + O(\varepsilon)$ uniformly.

Uniform sampling and the exact equilibrium give $\Gamma_\varepsilon \ge c\varepsilon^2$ for some $c > 0$. Conversely, changing a least-sampled entry (k, l) by $O(\varepsilon)$ invalidates any acceptable answer: choose the inequality indexed by $i = k$, $j \ne l$, whose coefficient on that entry is y_l, bounded away from zero. The perturbation remains admissible. Thus

$$\inf_{B \in \mathcal{B}_z} d_w(A, B) \le C_A \varepsilon^2 \min_{kl} w_{kl}.$$

The allocation supremum can therefore be restricted to a fixed compact subset of Δ_9°.

On this subset, the nearest weighted projection onto each invalidating hyperplane is $O(\varepsilon)$, hence admissible. Its exact squared-distance formula, (4), and uniform convergence of the coefficients imply

$$\lim_{\varepsilon \downarrow 0} \frac{\Gamma_\varepsilon}{\varepsilon^2} = \sup_{w \in \Delta_9^\circ} \max_{(a,b) \in K} \min_{ij} \frac{(1 - a_i - b_j)^2}{2\sigma_{ij}(w)^2}. \tag{5}$$

For fixed w, finite transportation duality gives

$$F_{p,q}(w) = \min_{u_i + v_j \ge \sigma_{ij}(w)} (p^\top u + q^\top v).$$

If $a_i + b_j + R\sigma_{ij} \le 1$, averaging against any coupling gives $RF_{p,q}(w) \le 1$. Conversely, for optimal dual potentials u, v, set

$$a_i = \frac{p^\top u - u_i}{F_{p,q}(w)}, \qquad b_j = \frac{q^\top v - v_j}{F_{p,q}(w)}.$$

These belong to K and attain $R = 1/F_{p,q}(w)$. Equations (2) and (5) prove (1). Every allocation coordinate appears with a nonzero coefficient in some c^{ij}; the product coupling gives that pair positive mass. Hence $F_{p,q}(w) \to \infty$ at the allocation boundary, proving attainment and positivity.

Each σ_{ij}, and therefore $F_{p,q}$, is convex in w. For uniform p, q, independent row and column permutations preserve F; averaging allocations proves uniform sampling optimal. There every $\sigma_{ij} = 2$, so $C = 8$.

For arbitrary p, q, take $w^0_{kl} = p_k q_l$. Then

$$\sigma_{ij}(w^0)^2 = p_i^{-1} + q_j^{-1} - 2, \qquad \sum_{ij} \pi_{ij}\sigma_{ij}(w^0)^2 = 4.$$

Jensen gives $C \le 8$. To prove strictness, set

$$h_{kl} = p_k q_l (p_k^{-1} + q_l^{-1} - 6), \qquad \sum_{kl} h_{kl} = 0.$$

Direct differentiation, uniformly over $\pi \in \Pi(p, q)$, gives

$$\sum_{ij} \pi_{ij}\sigma_{ij}(w^0 + th)^2 = 4 - t\left(\sum_i p_i^{-1} + \sum_j q_j^{-1} - 18\right) + O(t^2).$$

The parenthesis is strictly positive unless both strategies are uniform. Small positive t and Jensen therefore give $C < 8$.

Finally, A_r has the stated equilibrium by direct equalization. Under uniform sampling, $\sigma_{11} = 6r$, every $\sigma_{ij} \le 3\sqrt{2}$, and every coupling has $\pi_{11} \ge 1 - 4r$. Thus

$$\min_w F_{p_r, q_r}(w) \le (6 + 12\sqrt{2})r.$$

For every allocation, Cauchy–Schwarz gives $\sigma_{11} \geq 4r$; the diagonal coupling consequently gives

$$\min_{w} F_{p_r,q_r}(w) \geq 4r(1-2r).$$

This proves the asserted $\Theta(r^2)$ separation. For maximum-norm payoff estimation, centering at the true matrix optimizes each coordinate margin, and the corresponding local information is

$$\max_{w\in\Delta_9} \min_{kl} \frac{w_{kl}\varepsilon^2}{2} = \frac{\varepsilon^2}{18}.$$

Applying the same statistical reduction proves the comparison. With accuracy $\varepsilon/2$ in place of ε, the local information is $\varepsilon^2/72$. □

References

[1] Poiani, R., Bernasconi, M., and Celli, A. (2026). Pure Exploration with Infinite Answers. *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics*, PMLR 300, 2989–2997.

[2] Kaplansky, I. (1945). A Contribution to Von Neumann's Theory of Games. *Annals of Mathematics* 46(3), 474–479.